Name: [Your Name]
Lab Program Number: 12
Date: [Submission Date]

# Title: Understanding and Implementing 2D Reflection

## Theory

### *Reflection Along X-axis*

Reflection about the x-axis flips an object vertically, mirroring it across the x-axis. Each point (x,y) is transformed to (x,-y).

**Mathematical Representation (X-axis):**

x' = x
y' = -y

**Matrix Representation (X-axis):**

[x' y' 1] = [1 0 0; 0 -1 0; 0 0 1] · [x y 1]

### *Reflection Along Y-axis*

Reflection about the y-axis flips an object horizontally, mirroring it across the y-axis. Each point (x,y) is transformed to (-x,y).

**Mathematical Representation (Y-axis):**

x' = -x
y' = y

**Matrix Representation (Y-axis):**

[x' y' 1] = [-1 0 0; 0 1 0; 0 0 1] · [x y 1]

**2D Reflection Algorithm:**

1. Start
2. Choose axis of reflection (x or y)
3. For each point (x,y):
- If x-axis: (x',y') = (x,-y)
- If y-axis: (x',y') = (-x,y)
4. Update object coordinates
5. Stop

### *Program for Reflection Along X-axis*

```
#include <stdio.h>
#include <graphics.h>

void drawTriangle(int x1, int y1, int x2, int y2, int x3, int y3) {
    line(x1, y1, x2, y2);
    line(x2, y2, x3, y3);
    line(x3, y3, x1, y1);
```

```
}

void reflectX(int *x, int *y, int y_axis) {
    *y = 2 * y_axis - *y; // reflection about horizontal line y = y_axis
}

int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");

    int x1 = 100, y1 = 100;
    int x2 = 200, y2 = 100;
    int x3 = 150, y3 = 200;

    // Draw original triangle
    drawTriangle(x1, y1, x2, y2, x3, y3);

    int y_axis = 300; // reflection line y = 300 (can be adjusted)

    // Reflect about x-axis (horizontal line)
    reflectX(&x1, &y1, y_axis);
    reflectX(&x2, &y2, y_axis);
    reflectX(&x3, &y3, y_axis);

    // Draw reflected triangle
    drawTriangle(x1, y1, x2, y2, x3, y3);

    getch();
    closegraph();
    return 0;
}
```

## Output (Reflection Along X-axis)

**Output Description:**
The program displays a triangle before and after reflection about the x-axis. The original triangle is drawn in one color, and the reflected triangle is drawn in another. The output clearly labels:

**Object:** Original triangle
**Image:** Reflected triangle (mirrored vertically)
**Quadrant:** Triangles shown across quadrants separated by the x-axis.

(Attach a printout of your graphics output here, with clear labels.)

## *Program for Reflection Along Y-axis*

```
#include <stdio.h>
#include <graphics.h>

void drawTriangle(int x1, int y1, int x2, int y2, int x3, int y3) {
    line(x1, y1, x2, y2);
    line(x2, y2, x3, y3);
    line(x3, y3, x1, y1);
}

void reflectY(int *x, int *y, int x_axis) {
    *x = 2 * x_axis - *x; // reflection about vertical line x = x_axis
}

int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");

    int x1 = 100, y1 = 100;
    int x2 = 200, y2 = 100;
    int x3 = 150, y3 = 200;

    // Draw original triangle
    drawTriangle(x1, y1, x2, y2, x3, y3);

    int x_axis = 300; // reflection line x = 300 (can be adjusted)

    // Reflect about y-axis (vertical line)
    reflectY(&x1, &y1, x_axis);
    reflectY(&x2, &y2, x_axis);
    reflectY(&x3, &y3, x_axis);

    // Draw reflected triangle
    drawTriangle(x1, y1, x2, y2, x3, y3);

    getch();
    closegraph();
    return 0;
}
```

## Output (Reflection Along Y-axis)

**Output Description:**
The program displays a triangle before and after reflection about the y-axis. The original triangle is drawn in one color, and the reflected triangle is drawn in another. The output clearly labels:

**Object:** Original triangle
**Image:** Reflected triangle (mirrored horizontally)
**Quadrant:** Triangles shown across quadrants separated by the y-axis.

(Attach a printout of your graphics output here, with clear labels.)

## Conclusion

This lab successfully demonstrated the implementation of 2D reflection in Computer Graphics. Reflection about the x-axis and y-axis were both implemented, showing how objects can be mirrored across vertical and horizontal axes. The mathematical and matrix representations were validated through graphical output, confirming the correctness of the transformations. This exercise reinforced the concept of symmetry in 2D transformations and their practical applications in graphics.

## References

https://drive.google.com/file/d/1TGVqe6k75SXQRoCdWrzK2Za6gfhrO-BY/view?usp=sharing